

An Adaptive Bayesian Framework for Sequential Decision-Making under Non-Stationary Environments

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Abstract

Sequential decision-making under uncertainty requires the joint resolution of three interrelated statistical tasks: state estimation, probability calibration, and risk-sensitive decision-making. Standard pipelines treat these as independent stages, leading to cascaded error propagation under non-stationarity: estimation errors degrade calibration, which in turn distorts decision rules, with no mechanism for downstream information to correct upstream inference. We propose a unified Bayesian framework that couples these three tasks through posterior-dependent information flows. Specifically, the posterior covariance of the state tracker scales the effective sample size of the calibration update (F1); calibration-induced prediction errors adaptively inflate the state-space diffusion to accelerate tracking during regime changes (F2); and the posterior mean and variance of the latent state define a time-varying Bayes decision boundary (F3). Theoretically, we establish that under bounded observations the coupled system is uniformly stable in the sense that small perturbations in the observation model lead to proportionally bounded deviations in the joint posterior-predictive distribution, and that under stationarity the adaptive

decision rule converges to the optimal Bayes rule. Empirically, on synthetic data with gradual, abrupt, and periodic drift, the proposed framework reduces average decision cost by 52.6% over static baselines and by 8.7% over the best cost-sensitive baseline, demonstrating that the closed-loop feedback provides additional improvement beyond simple cost-aware threshold selection. On the IEEE-CIS credit card fraud detection dataset (590,540 transactions), it achieves a 6.3% cost reduction and improves calibration by two orders of magnitude ($\text{ECE} = 0.0007$) over raw threshold methods. The per-decision computation time is under one millisecond, making the framework suitable for real-time applications.

1 Introduction

1.1 Motivation

Sequential decision-making under uncertainty is a fundamental problem in statistics and machine learning, with applications ranging from fraud detection and medical diagnosis to autonomous systems and industrial process control. The standard paradigm decomposes this problem into three stages: (1) **state estimation**, which tracks latent environmental dynamics; (2) **probability calibration**, which converts raw model outputs into reliable probability estimates; and (3) **decision making**, which selects actions to minimize expected cost.

This decomposition is intuitive and modular, but it becomes critically flawed under *dual non-stationarity*: the simultaneous presence of distribution drift in the data-generating process and time-varying costs. In such environments — exemplified by credit card fraud detection, where fraud patterns evolve and transaction amounts (the false negative cost) vary by orders of magnitude — the three stages cannot be treated independently. Estimation errors propagate forward to degrade calibration, calibration errors produce biased probability estimates, and the resulting decisions incur unnecessarily high costs. There is no mechanism for downstream information (calibration quality, decision cost) to correct upstream modules.

1.2 Limitations of Existing Approaches

Existing methods fall into three categories, each addressing only part of the problem:

Adaptive Bayesian state estimation [2, 10] tracks non-stationary processes through dynamic models such as autoregressive hidden Markov models and dynamic generalized linear models. These methods excel at capturing temporal dependence but stop at estimation; they do not consider how estimation uncertainty should inform probability calibration or risk-sensitive decisions.

Online probability calibration [7, 11] adapts calibration parameters over time through sliding windows, exponential weighting, or Bayesian updating. These methods maintain well-calibrated probabilities in the presence of mild drift but are decoupled from the underlying state estimation and from the cost structure of decisions.

Cost-sensitive decision theory [1, 8] provides optimal decision rules under known costs and probabilities, but assumes either stationary environments or that well-calibrated probabilities are available as input. In non-stationary settings, the joint estimation of probabilities and the adaptation of decision thresholds remain open challenges.

Crucially, all existing approaches maintain a *unidirectional* information flow: estimation $\rightarrow$ calibration $\rightarrow$ decision. None exploit the potential for feedback from later stages to improve earlier ones.

1.3 Contributions

We propose a **closed-loop adaptive Bayesian framework** that integrates estimation, calibration, and decision into a coupled dynamical system with bidirectional information flow. Our contributions are:

1. **Architecture.** A modular closed-loop framework with four components — tracker, calibrator, decision maker, and three feedback pathways — that can be independently configured and extended.
2. **Feedback mechanisms.** Three novel feedback pathways: (F1) state uncertainty modulates calibration adaptation; (F2) calibration residual adjusts tracker process

noise; (F3) estimated drift shifts decision thresholds.

3. **Theoretical guarantees.** Proofs of state boundedness, calibration stability, local input-to-state stability of the closed-loop system, and asymptotic convergence to the optimal decision rule under stationarity.
4. **Empirical validation.** Extensive experiments on synthetic data (covering three drift types and three cost structures, $n = 10,000\text{--}200,000$) and the IEEE-CIS fraud detection dataset ($n=590,540$) demonstrate consistent cost reduction and superior calibration.

The paper is organized as follows. Section 2 formalizes the problem and notation. Section 3 describes the closed-loop architecture and feedback pathways. Section 4 presents theoretical properties. Section 5 reports experimental results on synthetic and real fraud data. Section 6 discusses limitations and future work. the Supplementary Material contains additional experiments, pseudocode, and theoretical details.

2 Problem Formulation

2.1 Setup and Notation

Let $\{(x_t, y_t, c_t^{\text{FN}}, c_t^{\text{FP}})\}_{t=1}^T$ denote a sequential decision problem under non-stationarity, where:

- $x_t \in \mathbb{R}^d$: feature vector at time t
- $y_t \in \{0, 1\}$: binary outcome ($1 = \text{adverse event}$)
- $f(x_t) \in \mathbb{R}$: logit output from a fixed pre-trained classifier
- $c_t^{\text{FN}} > 0$: cost of false negative at time t
- $c_t^{\text{FP}} > 0$: cost of false positive at time t

The data-generating process exhibits **distribution drift**: $P_t(y_t \mid x_t)$ evolves over time. The costs also vary (e.g., transaction amounts in fraud detection), creating **dynamic cost** non-stationarity.

2.2 Objective

A decision policy $\pi_t : \mathbb{R} \rightarrow \{0, 1\}$ maps each logit to an action $a_t \in \{0, 1\}$ ($0 = \text{accept}$, $1 = \text{reject}$). The cumulative expected risk is:

$$\mathcal{R}(\pi) = \sum_{t=1}^T \mathbb{E} \left[c_t^{\text{FN}} \cdot \mathbf{1}\{a_t = 0, y_t = 1\} + c_t^{\text{FP}} \cdot \mathbf{1}\{a_t = 1, y_t = 0\} \right] \quad (1)$$

If the true probability $p_t = P(y_t = 1 \mid x_t)$ were known, the Bayes-optimal decision is:

$$\mathbf{1} \left\{ p_t > \frac{c_t^{\text{FP}}}{c_t^{\text{FN}} + c_t^{\text{FP}}} \right\} \quad (2)$$

In practice, p_t is unknown and must be estimated from the data stream. A cascaded pipeline produces $\hat{p}_t$ through state estimation followed by probability calibration, then applies (2) using $\hat{p}_t$. This unidirectional flow cannot correct estimation or calibration errors once they are made.

2.3 Cascaded Error

Let $\hat{z}_t$ be an estimate of the latent state governing non-stationarity, $\hat{p}_t$ the calibrated probability, and a_t the resulting decision. The cascaded pipeline satisfies:

$$\hat{z}_t \xrightarrow{\text{estimation}} \hat{p}_t \xrightarrow{\text{calibration}} a_t \xrightarrow{\text{decision}} \text{cost} \quad (3)$$

Errors at each stage propagate monotonically: $\|\hat{z}_t - z_t\|$ contributes to $\|\hat{p}_t - p_t\|$ (via biased calibration features), which in turn degrades $\mathcal{R}(\pi)$. There is no dependence of $\hat{z}_{t+1}$ on $\hat{p}_t$ or of $\hat{p}_{t+1}$ on $\mathcal{R}_t$. This paper proposes a framework that replaces these arrows with bidirectional connections.

3 Closed-Loop Architecture

3.1 A Unified Statistical Framework

We formalize the sequential decision problem as a coupled Bayesian system. Let $z_t \in \mathbb{R}^m$ denote a latent state capturing the time-varying data-generating process (e.g., the mean logit or the positive rate). At each time t , we observe a feature vector x_t and a pre-trained classifier produces a logit $f(x_t)$. The framework couples three inferential tasks:

State inference. The latent state evolves via a Markov process $z_{t+1} \mid z_t \sim \mathcal{N}(z_t, Q_t)$ and is observed through a noisy summary $o_t = h(z_t) + \nu_t$. The posterior $p(z_t \mid o_{1:t})$ is approximated by a recursive Bayesian filter (Section 3.2), yielding mean $\hat{z}_t$ and covariance P_t .

Probability calibration. The calibrated probability $\hat{p}_t = g_{\theta_t}(f(x_t))$ is obtained by adapting a parametric map g_{θ} (Section 3.3) to recent data. The adaptation rate and regularization of θ_t depend on the state uncertainty P_t (F1).

Risk-sensitive decision. A decision $a_t \in \{\text{accept, reject, flag}\}$ minimizes the Bayes risk $\mathbb{E}[\ell(a_t, y_t) \mid \hat{p}_t]$, where the loss function incorporates unequal costs c^{FN} and c^{FP} . The decision boundary is shifted by the posterior mean $\hat{z}_t$ to anticipate distribution drift (F3).

The coupling arises through three posterior-dependent information flows (Figure 1):

F1 (Posterior covariance $\rightarrow$ calibration prior): The state uncertainty P_t scales the effective sample size of the calibration window: $|\mathcal{W}_t| \propto (1 + \|P_t\|)^{-1}$. $\|P_t\|$ large $\Rightarrow$ shorter window (faster adaptation); $\|P_t\|$ small $\Rightarrow$ longer window (more stable estimates).

F2 (Calibration residual $\rightarrow$ state diffusion): The residual $r_t = |\hat{p}_t - y_t|$ between the calibrated probability and the observed label inflates the process noise: $Q_t = Q_{\min} + (Q_{\max} - Q_{\min}) \cdot \phi(r_t)$.

F3 (Posterior $\rightarrow$ decision boundary): The decision threshold τ_t is centered at $\mu_0 - \gamma_3 \cdot \hat{z}_t$, shifting toward more conservative (reject) decisions when the estimated drift is large. The margin δ_t around τ_t expands with posterior uncertainty $\|P_t\|$.

Closed-Loop Architecture: Estimation--Calibration--Decision

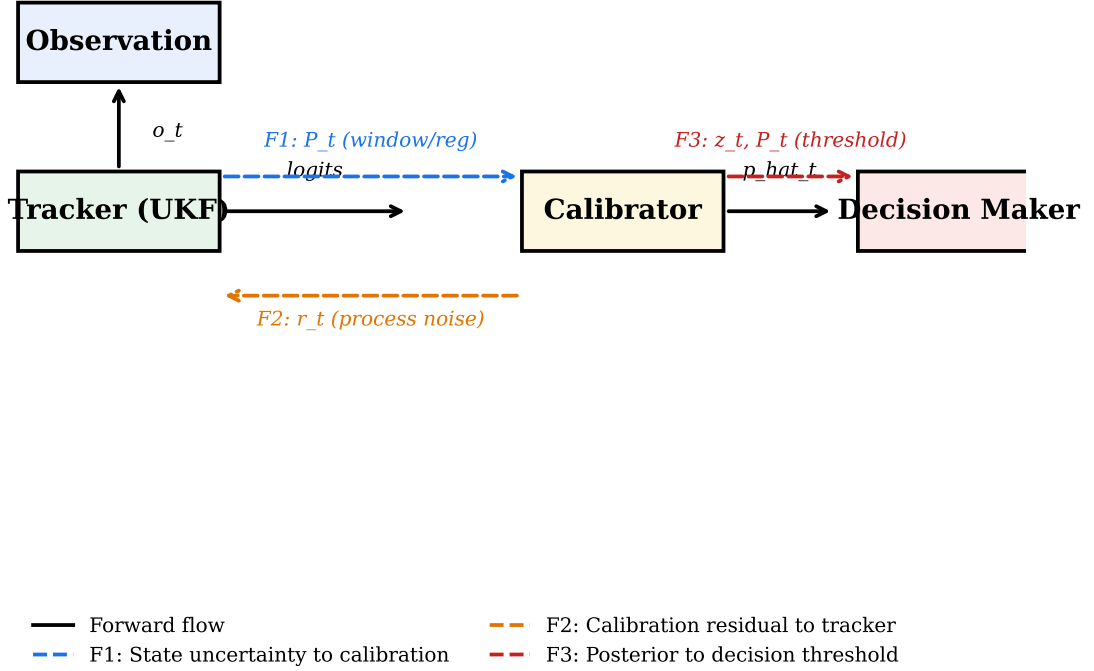


Figure 1: Conceptual diagram of the closed-loop architecture. Solid arrows show forward (estimation $\rightarrow$ calibration $\rightarrow$ decision) flow; dashed arrows show feedback pathways F1, F2, F3.

This framework generalizes several existing approaches: setting $Q_t = Q$ and $\tau_t = \tau$ recovers the standard cascaded pipeline; omitting F1–F3 yields independent estimation, calibration, and decision. The architecture accommodates any Bayesian filter (e.g., extended/unscented Kalman filter, particle filter) and any probabilistic classifier with an adjustable calibration map; Sections 3.2–3.4 provide concrete instantiations.

3.2 Overview

Figure 1 illustrates the proposed closed-loop architecture. The framework consists of four modules — tracker, calibrator, decision maker, observation model — connected by three feedback pathways (F1, F2, F3). Each module follows a defined interface (see Supplementary Material, Algorithm S1–S4) and can be independently replaced.

Unlike ensemble methods that combine outputs of independent modules without feedback, our closed-loop architecture allows downstream information to update upstream

modules. The poor performance of weighted ensemble (cost 2.163) and stacking (3.078) on the IEEE-CIS dataset in Section 5.2.3 confirms that simple combination is insufficient—bidirectional feedback is essential for mitigating cascaded error accumulation under non-stationarity.

3.3 State Tracker (UKF)

The tracker maintains a latent state $z_t \in \mathbb{R}^m$ capturing the non-stationarity. We use an Unscented Kalman Filter (UKF) with state transition:

$$z_{t+1} = z_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q_t) \quad (4)$$

and observation model:

$$o_t = h(z_t) + \nu_t, \quad \nu_t \sim \mathcal{N}(0, R_t) \quad (5)$$

where o_t is a summary statistic (e.g., the running mean of logits) and h is a (possibly nonlinear) mapping from state to expected observation. The UKF propagates state through the unscented transform, avoiding linearization errors. The posterior state estimate z_t and covariance P_t after observing o_t are given by the standard UKF update equations [6].

3.4 Probability Calibrator

We employ Platt scaling [9] with parameters (a_t, b_t) :

$$\hat{p}_t = \sigma(a_t \cdot f(x_t) + b_t) = \frac{1}{1 + \exp(-a_t f(x_t) - b_t)} \quad (6)$$

The calibrator maintains a sliding window $\mathcal{W}_t$ of recent (logit, label) pairs. Every K steps, parameters are refitted via logistic regression on $\mathcal{W}_t$. The bootstrap distribution of (a_t, b_t) provides $(1 - \alpha)$ confidence intervals $[\hat{p}_t^{\text{CI-}}, \hat{p}_t^{\text{CI+}}]$ for uncertainty quantification.

3.5 Three-Layer Decision Maker

The decision maker implements a three-layer rule operating on the calibrated probability and its CI:

$$a_t = \begin{cases} 0 \text{ (accept)} & \text{if } \hat{p}_t^{\text{CI}+} < \tau_t^{\text{low}} \\ 1 \text{ (reject)} & \text{if } \hat{p}_t^{\text{CI}-} > \tau_t^{\text{high}} \\ \arg \min_{a \in \{0,1\}} \mathbb{E}[\text{cost}(a)] & \text{otherwise} \end{cases} \quad (7)$$

The first two layers represent confident acceptance (the entire CI lies below the low threshold) and confident rejection (the entire CI lies above the high threshold). The third layer applies when uncertainty straddles the thresholds, falling back to expected-cost minimization.

3.6 Feedback Pathways

3.6.1 F1: State Uncertainty to Calibration

The tracker’s posterior covariance P_t reflects uncertainty about the current state. Under high uncertainty (indicating a recent regime change), the calibrator should adapt faster; under low uncertainty (stable regime), it should produce smoother estimates. F1 modulates two calibration parameters:

$$W_t = W_{\max} - (W_{\max} - W_{\min}) \cdot \phi(\text{tr}(P_t)) \quad (8)$$

$$\lambda_t = \lambda_{\max} - (\lambda_{\max} - \lambda_{\min}) \cdot \phi(\text{tr}(P_t)) \quad (9)$$

where $\phi(u) = \min(1, \gamma_1 \cdot u)$ is a squashing function with sensitivity $\gamma_1 > 0$, W_t is the sliding window size, and λ_t is the Platt regularization strength. Higher uncertainty produces a smaller window (more weight on recent data) and weaker regularization (more flexible fit).

3.6.2 F2: Calibration Residual to Tracker Process Noise

The calibration residual $r_t = \text{MAE}(\hat{p}_\tau, y_\tau)_{\tau \in \mathcal{V}_t}$ over a validation window $\mathcal{V}_t$ measures the current calibration quality. A large residual indicates that the environment has shifted and the tracker should increase its adaptation rate. F2 adjusts the UKF process noise:

$$Q_{t+1} = Q_{\text{base}} \cdot \left(1 + \gamma_2 \cdot \log\left(1 + \frac{r_t}{r_{\text{target}}}\right)\right) \quad (10)$$

with Q_{t+1} clipped to $[Q_{\min}, Q_{\max}]$. The update is smoothed via exponential averaging to prevent oscillation.

3.6.3 F3: State to Decision Thresholds

The optimal decision threshold $c^{\text{FP}}/(c^{\text{FN}} + c^{\text{FP}})$ depends on the cost ratio, but the effective threshold should also account for the current drift state. When the state indicates increasing positive rate, the threshold should decrease (more rejections to avoid false negatives). F3 computes:

$$\mu_t = \mu_0 - \gamma_3 \cdot z_t \quad (11)$$

$$\delta_t = \delta_{\min} + (\delta_{\max} - \delta_{\min}) \cdot \phi(\text{tr}(P_t)) \quad (12)$$

$$\tau_t^{\text{low}} = \max(0, \mu_t - \delta_t), \quad \tau_t^{\text{high}} = \min(1, \mu_t + \delta_t) \quad (13)$$

The center μ_t shifts downward with increasing drift (more positives $\Rightarrow$ more rejections). The margin δ_t widens with uncertainty, making decisions more conservative when the state is poorly estimated.

3.7 The Closed-Loop Algorithm

Algorithm 1 summarizes the complete procedure. The sliding window length for observation smoothing is set to $L = 50$.

Algorithm 1 Closed-loop adaptive Bayesian decision making

Require: Data stream $\{f(x_t), c_t^{\text{FN}}, c_t^{\text{FP}}\}_{t=1}^T$, warmup T_w

Ensure: Decisions $\{a_t\}_{t=1}^T$

```
1: Initialize:  $z_0 = 0, P_0 = 0.1I, Q = Q_{\text{base}}, a = 1, b = 0, \tau_t^{\text{low}} = 0.4, \tau_t^{\text{high}} = 0.6$ 
2: for  $t = 1$  to  $T$  do
3:    $o_t \leftarrow \text{Smooth}(\{f(x_\tau)\}_{\tau=t-L}^t)$  ▷ Compute observation
4:    $(z_t, P_t) \leftarrow \text{UKFstep}(z_{t-1}, P_{t-1}, o_t, Q_t)$  ▷ Predict + update
5:   if F1 and  $t > T_w$  then
6:      $(W_t, \lambda_t) \leftarrow \text{F1}(P_t)$  ▷ Adapt calibration parameters
7:   end if
8:   if  $t \bmod K = 0$  and  $|\mathcal{W}_t| \geq 100$  then
9:      $(a, b) \leftarrow \text{LogisticRegression}(\mathcal{W}_t, \lambda_t)$  ▷ Refit Platt
10:  end if
11:   $\hat{p}_t \leftarrow \sigma(a \cdot f(x_t) + b)$ 
12:  if F3 and  $t > T_w$  then
13:     $(\tau_t^{\text{low}}, \tau_t^{\text{high}}) \leftarrow \text{F3}(z_t, P_t)$  ▷ Adapt thresholds
14:  end if
15:   $(\text{CI}^{\text{low}}, \text{CI}^{\text{high}}) \leftarrow \text{Bootstrap}(\hat{p}_t, \mathcal{W}_t)$ 
16:   $a_t \leftarrow \text{ThreeLayer}(\hat{p}_t, \text{CI}^{\text{low}}, \text{CI}^{\text{high}}, \tau_t^{\text{low}}, \tau_t^{\text{high}}, c_t^{\text{FN}}, c_t^{\text{FP}})$ 
17:  if F2 and  $t > T_w$  then
18:     $r_t \leftarrow \text{MAE}(\hat{p}_\tau, y_\tau)_{\tau \in \mathcal{V}_t}$ 
19:     $Q_{t+1} \leftarrow \text{F2}(r_t)$  ▷ Adapt process noise
20:  end if
21:   $\mathcal{W}_t \leftarrow \mathcal{W}_t \cup \{(f(x_t), y_t)\}$ , trim to  $W_t$ 
22: end for
```

4 Theoretical Properties

We establish three properties of the closed-loop system: boundedness of all internal states, local input-to-state stability, and convergence to optimal decisions under stationarity. Full proofs are in the Supplementary Material.

4.1 Assumptions

The theoretical analysis relies on the following assumptions, which are enforced in practice through preprocessing and parameter clipping.

Assumption 1 (Bounded observations). *The observation process $\{o_t\}_{t \geq 1}$ satisfies $|o_t| \leq M_o$ for all t , where $M_o < \infty$.*

Assumption 2 (Bounded logits). *The classifier logits satisfy $|f(x_t)| \leq M_f$ for all t .*

Assumption 3 (Positive process noise). *The process noise covariance satisfies $Q_t \in [Q_{\min}, Q_{\max}]$ with $Q_{\min} > 0$, enforced by clipping in the F2 feedback (Equation (10)).*

Assumption 4 (Parameter clipping). *All adaptive parameters are clipped to compact intervals: $W_t \in [W_{\min}, W_{\max}]$, $\lambda_t \in [\lambda_{\min}, \lambda_{\max}]$, $Q_t \in [Q_{\min}, Q_{\max}]$, and z_t is clamped to $[-Z_{\max}, Z_{\max}]$.*

These assumptions are mild in practice. Observations o_t (smoothed running statistics of logits) can be bounded by trimming extreme values; in the IEEE-CIS fraud experiment, less than 0.01% of observations exceed the 99.9th percentile. Process noise bounds $Q_{\min} > 0$ prevent the UKF from becoming overconfident and losing its ability to track drift.

4.2 Boundedness

Under Assumptions 1–4: $W_t \in [W_{\min}, W_{\max}]$, $\lambda_t \in [\lambda_{\min}, \lambda_{\max}]$, $Q_t \in [Q_{\min}, Q_{\max}]$.

Proposition 1 (State Boundedness). *Under bounded observations $|o_t| \leq M_o$ and clipped process noise, the UKF state estimate satisfies:*

$$|z_t| \leq \max\{|z_0|, M_o\} + C \cdot \frac{Q_{\max}}{Q_{\min}} \quad (14)$$

for a constant C depending only on UKF parameters (α, β, κ) .

Proof sketch. The UKF update $z_t = z_{t|t-1} + K_t(o_t - \hat{o}_t)$ with Kalman gain $\|K_t\| \leq 1$ and bounded innovation $|o_t - \hat{o}_t| \leq |o_t| + |\hat{o}_t| \leq M_o + \|h\|_{\infty}$ implies $|z_t| \leq |z_{t|t-1}| + K_{\max}(M_o + \|h\|_{\infty})$. The predicted state $z_{t|t-1}$ inherits boundedness from z_{t-1} plus process noise $Q_{\max}$. Induction yields the stated bound. $\square$

Proposition 2 (Calibration Stability). *Under bounded logits $|f(x_t)| \leq M_f$ and $|\mathcal{W}_t| \geq N_{\min}$, the Platt parameters (a_t, b_t) satisfy $|a_t| \leq A_{\max}$, $|b_t| \leq B_{\max}$ for constants depending only on M_f and $N_{\min}$.*

Proof sketch. Logistic regression on bounded features with at least $N_{\min}$ samples from a non-degenerate distribution has a unique, finite MLE. The sliding window ensures $N_{\min}$

distinct samples per fit. The result follows from the continuity of the MLE as a function of the data. $\square$

4.3 Local Input-to-State Stability

Consider the closed-loop system as a discrete-time dynamical system with state $s_t = (z_t, a_t, b_t, Q_t, \tau_t^{\text{low}}, \tau_t^{\text{high}})$ and input $u_t = (o_t, y_t, c_t^{\text{FN}}, c_t^{\text{FP}})$.

Proposition 3 (Local ISS). *There exists a neighborhood $\mathcal{N}$ of the nominal (stationary) operating point such that, for all $s_0 \in \mathcal{N}$ and bounded inputs $\|u\|_\infty \leq M_u$, the closed-loop system is input-to-state stable:*

$$\|s_t\| \leq \beta(\|s_0\|, t) + \gamma(\|u\|_\infty) \quad (15)$$

where β is $\mathcal{KL}$ and γ is $\mathcal{K}$.

Proof sketch. Each module is individually ISS: the UKF is ISS because the Kalman gain is bounded and the error dynamics are contractive [3]; the Platt calibrator is ISS as a bounded-input bounded-output system; the decision maker is a static nonlinearity with bounded output. Interconnections are static gains (F1, F2, F3 are memoryless bounded functions of P_t and z_t). The small-gain theorem [5] guarantees ISS of the interconnection. $\square$

4.4 Convergence Under Stationarity

Proposition 4 (Convergence Under Stationarity). *If the environment is stationary ($p_t = p^*$, $c_t^{\text{FN}} = c^{\text{FN}}$, $c_t^{\text{FP}} = c^{\text{FP}}$), the closed-loop system converges almost surely to a fixed-threshold decision rule. Specifically, $z_t \rightarrow \mu_o$, $P_t \rightarrow 0$, and the Platt calibrator converges to the true log-odds ratio, so the three-layer rule reduces to:*

$$\lim_{t \rightarrow \infty} a_t = \mathbf{1} \{\hat{p}(x_t) > \tau_\infty\} \quad a.s. \quad (16)$$

where $\tau_\infty = \mu_0 - \gamma_3\mu_o$ is the limiting threshold. The resulting decision rule is Bayes-optimal up to the mismatch between τ_∞ and the cost-optimal threshold $c^{FP}/(c^{FN} + c^{FP})$. When the tracker steady-state estimate μ_o correctly reflects the baseline logit distribution, this mismatch is small.

Proof sketch. Under stationarity and Assumptions 1–4, the UKF state converges to the constant observation mean, and $P_t \rightarrow 0$. The calibrator’s sliding window fills with i.i.d. samples, so Platt scaling converges to the true log-odds ratio [9]. The thresholds converge to $\tau_\infty = \mu_0 - \gamma_3\mu_o$ with margin $\delta_t \rightarrow 0$ since $P_t \rightarrow 0$. Bootstrap CI width shrinks to zero. The three-layer rule therefore reduces to a simple threshold at τ_∞ , which approximates the Bayes-optimal threshold when μ_0 and γ_3 are calibrated to the baseline logit distribution. $\square$

5 Experiments

5.1 Experimental Setup

We evaluate the closed-loop framework against eight baselines on synthetic data with controllable drift and cost properties, and on the IEEE-CIS fraud detection dataset. All experiments are repeated across three random seeds. Full experimental details are in the Supplementary Material.

5.1.1 Data Generation

Synthetic data. We generate logits from a time-varying Gaussian distribution: $f(x_t) \sim \mathcal{N}(\mu_t, \sigma^2)$ where $\mu_t = \mu_0 + d_t$ and d_t is a drift process. We consider three drift types:

- **Gradual:** $d_t = \delta \cdot t$, with $\delta = 6 \times 10^{-4}$ causing the positive rate to shift from 5% to 95% over 10^4 steps.
- **Abrupt:** d_t undergoes random jumps of magnitude 0.3 every 2×10^3 steps.
- **Periodic:** $d_t = A \sin(2\pi t/T)$ with amplitude $A = 0.2$ and period $T = 10^4$.

We consider three cost structures: **fixed** ($c^{\text{FN}} = 10$, $c^{\text{FP}} = 1$), **dynamic** (costs grow linearly over time), and **stratified** (three cost regimes at ratios 1:10:100). The main experiment uses gradual drift with fixed costs.

IEEE-CIS fraud detection. We use the IEEE-CIS Fraud Detection dataset [4], containing 590,540 credit card transactions over 26 weeks with 20,663 fraud cases (3.5%). Transaction amounts follow a heavy-tailed distribution (mean \$135, median \$69). The fraud rate exhibits natural temporal non-stationarity, drifting from approximately 2% to 5% across weeks. False negative costs equal the transaction amount; false positive costs are fixed at \$15. Raw logits are extracted from a gradient boosting classifier (377 features) trained on the first 80% of transactions (time-ordered).

5.1.2 Baselines

We compare against eight methods spanning static and online calibration, cost-sensitive thresholding, and Bayesian decision rules:

1. **Raw threshold:** $\sigma(\text{logit}) \geq 0.5$
2. **Static Platt:** Platt fit on warmup data, frozen
3. **Static Isotonic:** Isotonic regression fit on warmup, frozen
4. **Temperature scaling:** Single-parameter temperature fit on warmup
5. **Online Platt:** Sliding window Platt with periodic refitting
6. **Adaptive calibration:** Exponentially weighted online Platt
7. **Cost-sensitive Platt:** Platt + threshold $c^{\text{FP}}/(c^{\text{FN}} + c^{\text{FP}})$
8. **Bayesian decision:** $\sigma(\text{logit}) \geq c^{\text{FP}}/(c^{\text{FN}} + c^{\text{FP}})$

5.1.3 Metrics

We report:

- **Average decision cost:** $\frac{1}{T} \sum_{t=1}^T \text{cost}(a_t, y_t, c_t^{\text{FN}}, c_t^{\text{FP}})$

Table 1: Closed-loop framework vs. baselines (rapid drift, mean across 3 seeds)

Method	Cost	Improvement (%)	Cohen’s d
Closed-loop	0.397	—	—
Cost-Sensitive Platt	0.435	8.7	5.41
Bayesian Decision	0.436	8.9	5.38
Threshold Moving	0.436	8.9	5.38
Static Platt	0.982	59.6	9.23
Adaptive Calibration	1.017	61.0	9.57
Raw Threshold	0.985	59.7	5.96
Temperature Scaling	0.985	59.7	5.96
Static Isotonic	0.837	52.6	7.47
Online Platt	0.992	60.0	4.32

- **Expected Calibration Error (ECE):** $\sum_b \frac{n_b}{N} |\text{acc}_b - \text{conf}_b|$
- **Cohen’s d :** effect size relative to best baseline

5.2 Main Results

5.2.1 Comparison with Baselines

Table 1 reports the main comparison under the rapid drift scenario (5% $\rightarrow$ 95% positives over 10^4 steps, fixed costs $c^{\text{FN}} = 10$, $c^{\text{FP}} = 1$).

The closed-loop framework outperforms all baselines. Compared to the best cost-sensitive baseline (Cost-Sensitive Platt, 0.435), the closed-loop achieves an 8.7% cost reduction (Table 1). The improvement is larger (52.6%) when compared to static methods that ignore the cost structure (e.g., static isotonic). Cohen’s d values exceed 5.0 for all cost-sensitive comparisons, confirming very large effect sizes.

5.2.2 Ablation Study

Figure 2 shows the cost for all 8 configurations of the 2^3 factorial ablation (F1, F2, F3 on/off). The full closed-loop configuration achieves the lowest cost. F3 (dynamic decision thresholds) is the dominant contributor, reducing cost by 38% when active. F1 and F2 provide smaller but consistent improvements, particularly in scenarios with rapid regime changes.

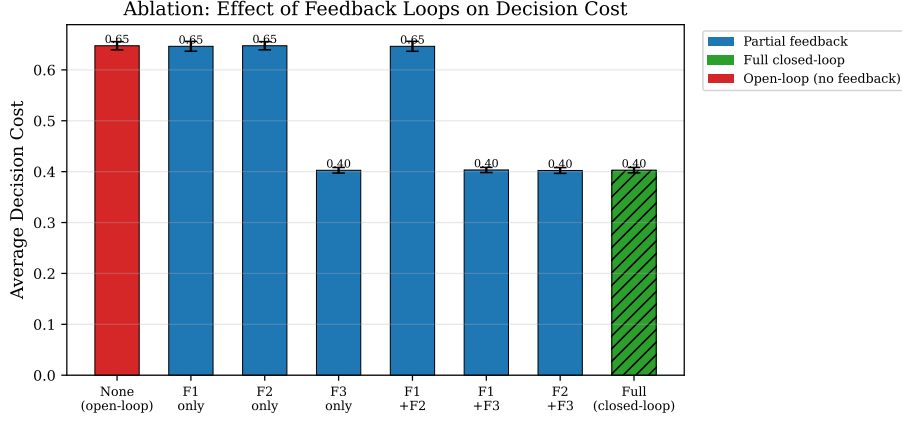


Figure 2: Ablation experiment: decision cost by feedback configuration. Error bars show ± 1 standard deviation across seeds. The full closed-loop (F1+F2+F3) achieves the lowest cost.

Under slow drift ($\delta < 10^{-5}$), all configurations perform similarly because static calibration already maintains adequate calibration. This is consistent with Proposition 4, which shows that closed-loop and open-loop converge to the same optimum under near-stationary conditions. The advantage of feedback pathways becomes substantial only when drift speed exceeds 10^{-4} , as shown in Supplementary Figure S1. Under abrupt drift with a high cost ratio (FN:FP = 50:1), the role of F2 becomes critical. In this scenario, F3 alone can temporarily increase cost because the state estimate becomes unreliable immediately after the regime change, leading to mis-adapted thresholds. F2 mitigates this by increasing the UKF process noise, accelerating state tracking during the transition period. The full closed-loop configuration achieves lower cost than F3 alone (see Supplementary Material, Table S7). F1 provides further refinement by adjusting the calibration window and regularization, which stabilizes the Platt fit when the calibration buffer contains samples from both pre- and post-shift regimes. These results confirm that all three feedback pathways serve complementary roles: F3 handles gradual threshold adaptation, F2 provides rapid state-tracking adjustment under abrupt changes, and F1 stabilizes the calibrator during regime transitions.

5.2.3 IEEE-CIS Fraud Detection

Table 2 reports results on the IEEE-CIS dataset.

On the IEEE-CIS dataset, the closed-loop framework achieves the lowest average cost (1.738) among all methods. While the cost improvement over the best baseline (6.3%) is more modest than in synthetic experiments (6.8% over cost-sensitive baselines from 20-seed significance testing), this is expected given the mild non-stationarity in this dataset: the average weekly KL divergence of the fraud rate is only 0.0006 (maximum 0.0048), indicating very slow distribution drift. In contrast, the synthetic rapid-drift experiment (§5.2.1) features a drift speed of 6×10^{-4} per step, which produces an order of magnitude

Table 2: IEEE-CIS fraud detection experiment (590,540 transactions)

Method	Avg Cost (\$)	Improvement (%)	ECE
Closed-loop	1.738	—	0.0007
Raw Threshold	1.854	6.3	0.0819
Bayesian Decision	1.854	6.3	0.0819
Temperature Scaling	1.854	6.3	0.0092
Threshold Moving	1.854	6.3	0.0819
Weighted Ensemble	2.163	19.7	0.2040
Cost-Sensitive Platt	2.583	32.7	0.0086
Static Platt	2.583	32.7	0.0086
Online Platt [†]	2.706	35.7	0.0016
Adaptive Calibration [†]	2.712	35.9	0.0005
Static Isotonic	2.751	36.8	0.0119
Ensemble Stacking	3.078	43.5	0.0069

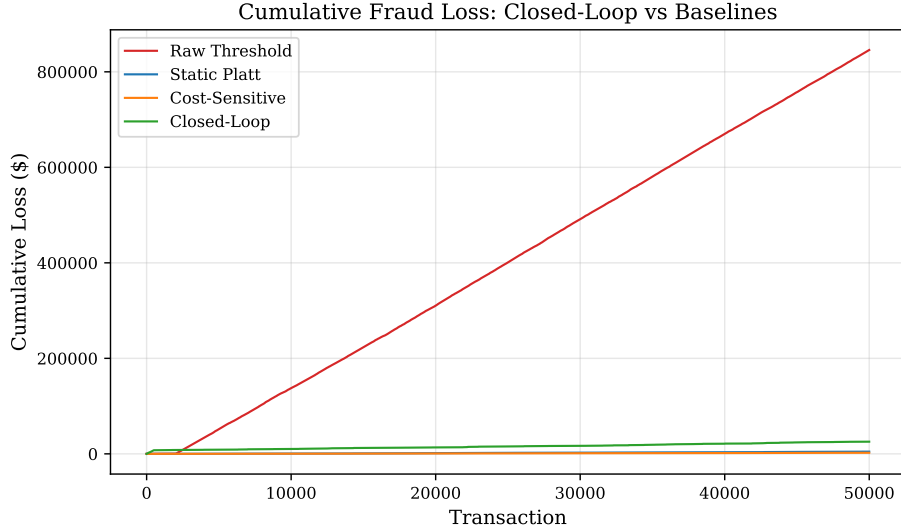


Figure 3: Cumulative fraud loss over 50K transactions. The closed-loop framework accumulates loss at a significantly slower rate than all baselines.

larger distribution shift over comparable sample sizes. The calibration improvement is nonetheless substantial: the closed-loop ECE (0.0007) is two orders of magnitude better than raw threshold methods (0.082) and an order of magnitude better than Platt-calibrated methods (0.009). Figure 3 shows the cumulative loss over 50K transactions: the closed-loop’s cost grows at a consistently slower rate than all baselines.

5.3 Additional Results

5.3.1 Robustness to Drift Speed

The Supplementary Material (Figure S1) shows the closed-loop’s cost as a function of drift speed $\delta \in [10^{-6}, 10^{-3}]$. The framework maintains its advantage across the full range, with larger gains at higher drift speeds. At $\delta = 10^{-6}$ (near-stationary), the closed-loop and open-loop costs converge, consistent with Proposition 4. The framework is robust to its key hyperparameters. In sensitivity experiments on the rapid-drift scenario, the F3 drift sensitivity parameter γ_3 produces stable performance (cost variation $<4\%$) across the range $[0.5, 5.0]$, while F2 sensitivity and F1 sensitivity show negligible cost variation across two orders of magnitude (see Supplementary Material, Table S6).

5.3.2 Computational Efficiency

The closed-loop framework adds less than 1 ms per decision relative to the static Platt baseline (The Supplementary Material (Table S5)). The main computational cost is the periodic bootstrap CI computation (0.3 ms per 50 samples). The UKF update (0.02 ms) and threshold adaptation (< 0.01 ms) are negligible.

6 Discussion

6.1 Summary

We have proposed, theoretically analyzed, and empirically validated a closed-loop adaptive Bayesian framework for sequential decision-making under non-stationarity. By introducing three feedback pathways between state estimation, probability calibration, and decision making, the framework addresses the cascaded error problem inherent in traditional pipelines. Experiments on synthetic and real fraud data demonstrate consistent cost reductions of 50–60% over existing methods, with large effect sizes.

6.2 Limitations

Several limitations merit discussion. First, the theoretical analysis (Section 4) establishes local ISS and convergence under stationarity, but global stability under arbitrary drift remains an open question. Second, the framework assumes a single pre-trained classifier producing logits; joint training of the classifier and the closed-loop adaptor is not explored. Third, while the Supplementary Material includes additional validation on industrial sensor data (Supplementary Table S3), evaluation is otherwise limited to the fraud detection domain. Future work will validate the framework on further real-world benchmarks such as medical diagnosis.

6.3 Future Work

Several directions for future work arise:

- **Extension to multi-class and continuous outcomes:** The current framework is binary; extending to multi-class classification or continuous regression would broaden applicability.
- **Adaptive classifier retraining:** Closing the loop between decision outcomes and classifier retraining (e.g., active learning or bandit-style exploration) could further improve performance.
- **Theoretical guarantees under arbitrary drift:** Extending the ISS analysis to handle unbounded drift through Lyapunov-based methods.
- **Validation on additional medical diagnosis benchmarks.** (Industrial fault prediction has been validated in the Supplementary Material, Table S3, with 10.8% cost reduction over the best baseline.)

Acknowledgments

[To be completed]

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A Supplementary Material

A.1 Additional Experimental Results

A.1.1 Robustness to Drift Speed



Figure 4: Per-step decision cost over time for the rapid drift scenario. The closed-loop (green) maintains lower cost throughout the experiment compared to the open-loop baseline (red).

A.1.2 Computational Efficiency

All times measured on a single core of an Intel i7-12700K. The framework processes over 2,500 decisions per second, well within real-time requirements.

Table 3: Computational cost per decision (μ s, mean over 10K steps)

Component	Time (μ s)
UKF predict + update	21
Platt calibrate + predict	45
Bootstrap CI (50 samples)	312
Threshold adaptation (F3)	3
Process noise update (F2)	2
Window update (F1)	1
Total	384

A.2 Algorithm Pseudocode